

PHASE 3: COMPLETE STEP-BY-STEP EXECUTION GUIDE

12 detailed steps — from prerequisites to Phase 4 preparation

BEFORE YOU START

Total time estimate:

Setup:	5 minutes
Execution:	2 minutes
Interpretation:	30 minutes

Total:	~40 minutes
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What you need:

- langmuir_predictions.csv (from Phase 2)
- Python 3.8+ installed
- scikit-learn, scipy, matplotlib, pandas, numpy
- phase3_residual_analysis.py script

STEP 1: VERIFY PHASE 2 OUTPUT FILE

What to do:

```
ls -lh data/langmuir_predictions.csv
# OR if in a different location:
ls -lh /path/to/langmuir_predictions.csv
```

What you should see:

```
-rw-r--r-- 1 user group 43K May 3 12:45 data/langmuir_predictions.csv
```

Quick sanity check:

```
python3 -c "
import pandas as pd
df = pd.read_csv('data/langmuir_predictions.csv')
print(f'Rows: {len(df)}')
print(f'Cols: {list(df.columns)}')
print(f'Residual range: {df.residual.min():.3f} to {df.residual.max():.3f}')
"
```

Expected:

```
Rows: 500
Cols: ['Run', 'pH', 'CO', 'Time', 'Dose', 'Temp', 'Flow',
       'Chloride', 'Hardness', 'Carbonate', 'NOM', 'Order',
       'q_removal', 'q_predicted', 'residual']
Residual range: -1.101 to 2.210
```

If file is missing:

```
# Copy from your Phase 2 output folder
cp /path/to/phase2_outputs/langmuir_predictions.csv data/
```

STEP 2: INSTALL DEPENDENCIES

```
pip install scikit-learn scipy matplotlib pandas numpy
```

Verify:

```
python3 -c "
import sklearn, scipy, matplotlib, pandas, numpy
print(f'sklearn: {sklearn.__version__}')
print(f'scipy:    {scipy.__version__}')
print(f'pandas:   {pandas.__version__}')
print('All OK!')
"
```

Expected:

```
sklearn: 1.x.x
scipy:    1.x.x
pandas:   2.x.x
All OK!
```

STEP 3: CREATE DIRECTORY STRUCTURE

```
mkdir -p data results
ls -la
```

Expected structure:

```
project/
  data/
    langmuir_predictions.csv  ← Phase 2 output
  results/                   ← Phase 3 will write here
    phase3_residual_analysis.py ← The script
```

STEP 4: RUN THE SCRIPT

```
python3 phase3_residual_analysis.py
```

Expected runtime: 1–2 minutes (most time is the Random Forest with 200 trees)

STEP 5: MONITOR STEP [1/7] — DATA LOADING

What you see:

```
=====
PHASE 3: RESIDUAL ANALYSIS & FEATURE ENGINEERING
Fluoride Adsorption on Coconut Husk Activated Carbon
=====
```

```
[1/7] Loading Phase 2 results...
Loaded 500 samples, 15 columns
q_removal range: 1.417 - 8.316 mg/g
Residuals: mean=-0.000000, std=0.5283 mg/g
Residuals: min=-1.1008, max=2.2100 mg/g
Skewness: 0.6008 | Kurtosis: 0.1135
```

What to check: - 500 samples loaded (not 499, not 501) - Residual mean 0 (should be $< 1e-6$ in magnitude) - Std 0.528 (matches Phase 2 RMSE) - Skewness positive (+0.60) = right-skewed as expected

If you see fewer rows: → Data file may be corrupted or truncated. Re-copy from Phase 2 output.

STEP 6: MONITOR STEP [2/7] — PATTERN ANALYSIS

What you see:

```
[2/7] Residual pattern analysis by factor...
pH pattern analysis:
pH Range    Mean Res    Std    Count    Pattern

pH 3-4      +0.248     0.492    84      ↑ underest.
pH 4-5      -0.525     0.290    83      ↓ overest.
pH 5-6      -0.117     0.394    84      random
pH 6-6.5    +0.471     0.448    41      ↑ underest.
pH 6.5-7    +0.649     0.502    42      ↑ underest.
pH 7-7.5    +0.208     0.418    41      ↑ underest.
pH 7.5-8    -0.176     0.271    42      random
pH 8-9      -0.182     0.336    83      random
```

```
Time pattern: {...} → No significant pattern
Dose pattern: {...} → Minor pattern at low doses
Temp pattern: {...} → No pattern (Arrhenius works)
```

```
Linear correlations with residuals (all ~zero = expected):
Max |corr|: 8.91e-16 (polynomial regression removed all linear signal)
```

```
Large residuals (|res| > 1.057 = 2):
Positive (model underestimates): 17 points (3.4%)
```

Negative (model overestimates): 1 points (0.2%)
Mean pH at large positive: 6.11 (near optimal 6.5-7)
Mean pH at large negative: 4.34 (slightly acidic 4-5)

What to check: - pH 6.5-7 shows +0.649 (biggest positive bar) - pH 4-5 shows -0.525 (biggest negative bar) - Time, Temp show near-zero means (no pattern) - Linear correlations all ~ machine epsilon ($\sim 1e-16$) - 17 large positive outliers at mean pH 6.11

The big finding: pH 6.5-7 has mean residual +0.649 — this is your ML goldmine.

STEP 7: MONITOR STEP [3/7] — FEATURE ENGINEERING

What you see:

```
[3/7] Engineering new features...
Original factors kept: 10
New features created: 28
Total features for ML: 38
Feature groups:
pH deviation:      5 features
Ion competition:   5 features
Equilibrium/load:  5 features
Log transforms:    5 features
Interactions:      5 features
Composite:         3 features
```

What to check: - 28 new features created - Total = 38 features

STEP 8: MONITOR STEP [4/7] — CORRELATIONS

What you see:

```
[4/7] Engineered feature correlations with residuals...
Top 10 features correlated with residuals:
1. pH_optimal          r = +0.4837
2. high_perf_flag      r = +0.3523
3. pH_gaussian         r = +0.2231
4. pH_abs_dev          r = -0.1998
5. optimal_pH_score    r = +0.1806
6. ion_pH_interaction  r = -0.1708
7. adsorbent_excess    r = -0.0376
8. CO_Dose_ratio       r = +0.0354
9. fouling_impact      r = +0.0205
10. ion_ratio          r = -0.0128
```

What to check: - pH_optimal has largest correlation (+0.48) - Top features are pH-related - Correlations are NOT near machine epsilon (unlike original factors)

Why pH_abs_dev shows negative correlation here:

pH_abs_dev = |pH - 6.5|

When pH is near optimal (6-7): pH_abs_dev is SMALL → residual is POSITIVE

When pH is far from optimal: pH_abs_dev is LARGE → residual is NEGATIVE

So higher pH_abs_dev → lower residual → negative correlation!

This still makes it USEFUL for ML - it's a valid signal.

STEP 9: MONITOR STEP [5/7] — TRAIN/TEST SPLIT

What you see:

```
[5/7] Preparing ML training dataset...
      Train set: 400 samples (80%)
      Test set: 100 samples (20%)
      Feature matrix: 38 features
      Target: residual (mean=0.0106, std=0.5416)
```

What to check: - 400 train + 100 test = 500 total - 38 features (10 original + 28 engineered)
- Target variable is 'residual' - mean 0 (slight deviation due to random split is normal)

STEP 10: MONITOR STEP [6/7] — QUICK RANDOM FOREST

What you see (takes ~60-90 seconds):

```
[6/7] Quick Random Forest - feature importance ranking...
      RF train R2 (on residuals): 0.9157
      RF test R2 (on residuals): 0.4697
      RF test RMSE: 0.3415 mg/g
```

Top 15 features:

Rank	Feature	Importance	Cum%
1	pH_abs_dev	0.2204	22.0%
2	pH_dev_sq	0.2078	42.8%
3	pH_gaussian	0.0994	52.8%
4	pH	0.0745	60.2%
5	pH_dev	0.0605	66.3%
...			

What to check: - Train R² > 0.85 (confirms features are useful) - Test R² > 0.35 (confirms generalisation exists) - Top 3 features are all pH-related - pH_abs_dev and pH_dev_sq are #1 and #2

Understanding the train-test gap (0.9157 vs 0.4697):

Gap = 0.45 → looks large but is EXPECTED because:

1. No hyperparameter tuning yet (Phase 4 will fix this)

2. Quick RF uses max_depth=8 → overfits
3. Residual prediction is inherently harder than q prediction
4. Phase 4 with CV and tuning will close this gap

Bottom line: test $R^2 = 0.47$ is still meaningful!

STEP 11: MONITOR STEP [7/7] — SAVING OUTPUTS

What you see:

```
[7/7] Saving results and creating visualisations...
      Saved: results/ml_training_data.csv ((400, 39))
      Saved: results/ml_test_data.csv    ((100, 39))
      Saved: results/feature_importance.csv ((38, 4))
      Saved: results/residual_analysis.json
      Saved: results/phase3_diagnostics.png (6-panel plot)
```

Then the final summary:

```
=====
PHASE 3 COMPLETE: RESIDUAL ANALYSIS & FEATURE ENGINEERING
=====
```

Key Findings:

- pH is the dominant residual driver
- pH 4-5: mean residual = -0.525 mg/g (model overestimates)
- pH 6.5-7: mean residual = +0.649 mg/g (model underestimates - ML goldmine)
- 28 new features engineered from 10 original factors
- Top feature: pH_abs_dev (importance=0.2204)
- Quick RF on residuals: train $R^2=0.9157$, test $R^2=0.4697$

Output Files:

```
results/ml_training_data.csv (400 samples × 38 features)
results/ml_test_data.csv    (100 samples × 38 features)
results/feature_importance.csv (38 features ranked)
results/residual_analysis.json (full metadata)
results/phase3_diagnostics.png (6-panel diagnostics)
```

STEP 12: VERIFY OUTPUT FILES

Check all 5 files created:

```
ls -lh results/
```

Expected:

```
-rw-r--r-- 1 user group 186K May 5 13:20 results/ml_training_data.csv
-rw-r--r-- 1 user group  47K May 5 13:20 results/ml_test_data.csv
```

```
-rw-r--r-- 1 user group 2K May 5 13:20 results/feature_importance.csv
-rw-r--r-- 1 user group 1K May 5 13:20 results/residual_analysis.json
-rw-r--r-- 1 user group 500K May 5 13:20 results/phase3_diagnostics.png
```

Verify CSV structure:

```
import pandas as pd, json

train = pd.read_csv('results/ml_training_data.csv')
test = pd.read_csv('results/ml_test_data.csv')
feat = pd.read_csv('results/feature_importance.csv')

print(f"Train: {train.shape} (expected 400 × 39)")
print(f"Test: {test.shape} (expected 100 × 39)")
print(f"Feat: {feat.shape} (expected 38 × 4)")
print(f"Train residual std: {train['residual'].std():.4f}")
print(f"Top feature: {feat.iloc[0]['feature']} ({feat.iloc[0]['importance']:.4f})")

with open('results/residual_analysis.json') as f:
    meta = json.load(f)
print(f"Top feature from JSON: {meta['key_findings']['top_feature']}")
print(f"pH 6.5-7 mean residual: {meta['key_findings']['pH_6p5_7_mean_residual']:.4f}")
```

Expected:

```
Train: (400, 39)
Test: (100, 39)
Feat: (38, 4)
Train residual std: 0.54xx
Top feature: pH_abs_dev (0.2204)
Top feature from JSON: pH_abs_dev
pH 6.5-7 mean residual: 0.6491
```

Open diagnostic plot:

```
# Linux/Mac
open results/phase3_diagnostics.png
# OR
display results/phase3_diagnostics.png
```

COMPLETE SUCCESS CHECKLIST

Setup: - Phase 2 CSV loaded (500 rows, 15 columns) - Dependencies installed (sklearn, scipy, matplotlib) - data/ and results/ directories exist - Script in working directory

Execution: - All 7 steps complete without errors - pH pattern confirmed (6.5-7 highest positive bar) - 38 features created (10 original + 28 engineered) - pH_optimal correlation > 0.40 - Top features are pH-based - Train $R^2 > 0.85$, Test $R^2 > 0.35$

Outputs: - ml_training_data.csv (400 × 39) - ml_test_data.csv (100 × 39) - fea-

ture_importance.csv (38 × 4) - residual_analysis.json - phase3_diagnostics.png (6 panels)

Understanding: - You know why pH is the dominant signal - You understand why linear correlations are all zero - You understand the train/test gap (overfitting = normal at Phase 3) - You know how Phase 4 will improve the RF results - You know the hybrid model path to R^2 0.94

IF ALL CHECKED: PHASE 3 COMPLETE!

TROUBLESHOOTING

Error	Cause	Fix
FileNotFoundError: data/langmuir_predictions.csv	Wrong path	<code>mkdir -p data && cp /path/to/file data/</code>
ModuleNotFoundError: sklearn	Not installed	<code>pip install scikit-learn</code>
TypeError: cut() got unexpected argument 'q'	Older script version	Use <code>qcut()</code> instead of <code>cut()</code> for quartiles
Train $R^2 < 0.70$	Features not computed right	Print first 5 rows of engineered features
Test $R^2 < 0.20$	Severe overfitting	Reduce <code>max_depth</code> to 4 in Quick RF
Missing output files	Permission error	<code>chmod 755 results/</code> and re-run
Plot looks empty	Matplotlib backend	Add <code>matplotlib.use('Agg')</code> before imports

PREPARING FOR PHASE 4

Files you'll use:

results/ml_training_data.csv → X_train + y_train
results/ml_test_data.csv → X_test + y_test
results/feature_importance.csv → Feature selection guidance
results/residual_analysis.json → Baseline metrics reference

What Phase 4 does:

1. Loads ml_training_data.csv and ml_test_data.csv
2. Trains Random Forest with 5-fold CV and hyperparameter tuning
3. Trains XGBoost with 5-fold CV
4. Trains MLP Neural Network
5. Selects best model (expected winner: XGBoost)
6. Produces residual predictions for all 500 points

7. Passes to Phase 5 for hybrid combination

Expected Phase 4 improvements:

Phase 3 Quick RF test R^2 : 0.4697

Phase 4 tuned RF test R^2 : 0.65-0.70

Phase 4 XGBoost test R^2 : 0.68-0.75 ← likely winner

Phase 4 MLP test R^2 : 0.55-0.68

Phase 5 projection:

If Phase 4 achieves $R^2(\text{residuals}) = 0.72$:

Additional variance = $0.72 \times 18.42\% = 13.3\%$

Hybrid $R^2 = 0.8158 + 0.133 = 0.949 \rightarrow \text{EXCEEDS } 0.94!$

Phase 3 Complete! You're 50% through the 8-phase project.

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